
Decomposing Uncertainty in Probabilistic Knowledge Graph Embeddings: Why Entity Variance Is Not Enough

Abstract

Probabilistic knowledge graph embeddings use learned variances to quantify epistemic uncertainty, but these variances are *relation-agnostic*—an entity receives identical uncertainty regardless of relational context. This conflates two distinct OOD phenomena: *emerging entities* (rare, poorly-learned) and *novel relational contexts* (familiar entities in unobserved relationships). We prove that, under empirically validated frequency-matching assumptions, any uncertainty estimator using only entity-level statistics achieves near-random detection on novel contexts, explaining why existing methods score 0.99 AUROC on random corruptions but only 0.38–0.63 on temporal distribution shift. We decompose uncertainty into complementary *semantic* (entity variance) and *structural* (entity-relation co-occurrence) components, and characterize their distinct failure modes across OOD types. The decomposition achieves 0.86–0.99 AUROC on temporal OOD across three static benchmarks and 0.99 on ICEWS14 (a temporal KG with ground-truth timestamps), substantially improving over relation-agnostic baselines (0.38–0.63).

1 INTRODUCTION

Knowledge graph (KG) embeddings enable efficient reasoning over structured knowledge [Bordes et al., 2013, Trouillon et al., 2016]. As KGs evolve—new entities emerge, existing entities appear in novel relationships—robust uncertainty quantification becomes critical.

The Relation-Agnostic Limitation. Recent probabilistic methods learn distributions over embeddings [Chen et al., 2019, 2021, Choudhary et al., 2021], where rare entities have

higher variance. But these variances are *relation-agnostic*: entity e receives a single variance σ_e^2 regardless of relational context. Consider e appearing in 1000 triples with relations $\{r_1, \dots, r_{10}\}$ but never with r_{11} . The model assigns low variance (well-observed), yet queries $(?, r_{11}, e)$ involve genuinely uncertain predictions.

This causes systematic failures (Figure 1): score-based methods achieve 0.99 AUROC detecting random corruptions (implausible triples) but only 0.38–0.60 on temporal OOD (emerging entities and novel relational contexts that simulate temporal distribution shift).

Two OOD Types. We identify two distinct scenarios: (1) **Emerging entities:** New/rare entities with poorly-constrained embeddings—high variance correctly signals uncertainty. (2) **Novel contexts:** Well-observed entities in unobserved relations—variance is low (falsely confident), but the entity-relation pair was never trained.

Contributions. We identify a systematic limitation in probabilistic KG embeddings: learned variances are relation-agnostic despite training on data containing entity-relation co-occurrence patterns. This failure reveals a fundamental mismatch between standard training objectives (link prediction accuracy) and OOD detection requirements (capturing structural observation patterns).

Our contributions are: (1) **Problem identification and impossibility result:** We identify a systematic blind spot in probabilistic KG embeddings—the inability to detect novel relational contexts—and prove that *any* relation-agnostic uncertainty estimator achieves near-random detection on this category (Theorem 1). (2) **Uncertainty decomposition framework:** We formalize complementary semantic and structural uncertainty signals and characterize their failure modes by OOD type (semantic fails on novel contexts; structural degrades on emerging entities; Theorem 2). This decomposition is the core contribution; the combination mechanism (CAGP) is intentionally minimal. (3) **Empirical validation:** We demonstrate this limitation persists across

existing methods (0.38–0.63 AUROC on temporal shift despite 0.99 on random corruptions), and that the decomposition achieves 0.86–0.99 AUROC across four benchmarks.

2 RELATED WORK

Probabilistic KG Embeddings. UKGE [Chen et al., 2019] associates confidence scores with triples; BEUrRE [Chen et al., 2021] uses box embeddings where volume indicates uncertainty; GP-KGE [Choudhary et al., 2021] learns Gaussian distributions over entities. All share the limitation we identify: learned variances are relation-agnostic. Despite being trained on data containing entity-relation co-occurrence patterns, these methods do not autonomously learn relation-specific uncertainty, necessitating our explicit decomposition framework. While relation-aware scoring functions exist (R-GCN [Schlichtkrull et al., 2018], CompGCN [Vashishth et al., 2020]), these condition *predictions* on relations but still produce relation-agnostic uncertainty estimates; our framework addresses uncertainty decomposition orthogonally to scoring architecture.

OOD Detection. MC Dropout [Gal and Ghahramani, 2016], Deep Ensembles [Lakshminarayanan et al., 2017], and Energy-based methods [Liu et al., 2020b] capture model uncertainty but not structural observation patterns. Simple Neural Gaussian Process (SNGP) [Liu et al., 2020a] estimates distance to training data but remains relation-agnostic. Graph Posterior Networks [Stadler et al., 2021] propagate uncertainty through graphs but treat edges homogeneously. Classical feature-space methods (Mahalanobis distance [Lee et al., 2018], energy scores) assume continuous feature geometry; KGs’ discrete combinatorial structure motivates our coverage-based alternative.

Temporal Knowledge Graphs. Temporal KG methods extend embedding models with time-aware scoring functions for link prediction over evolving graphs [García-Durán et al., 2018, Lacroix et al., 2020, 2021]. These methods model *when* facts hold but do not address uncertainty quantification or OOD detection—the problem we target. Our simulated temporal splits test whether methods can flag unseen entity-relation patterns, complementing temporal KG research.

Positioning. Prior work treats uncertainty as a single quantity. We show KG uncertainty decomposes into semantic (entity-level) and structural (relation-specific) components, with theoretical guarantees on when each is necessary (Theorem 2). Our decomposition mirrors the aleatoric/epistemic distinction in Bayesian deep learning [Kendall and Gal, 2017]: semantic uncertainty captures epistemic uncertainty about entity embeddings, while structural uncertainty captures the absence of training signal for specific entity-

relation combinations. Recent work has advanced KG uncertainty from complementary angles: PAC-Bayesian generalization bounds for KGE [Lee et al., 2024] and conformal prediction for uncertain KGs [Zhu et al., 2025]. These address generalization and calibration, respectively, while our work targets the orthogonal problem of *decomposing* uncertainty into relation-aware components for OOD detection.

3 BACKGROUND

Knowledge Graphs. A knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ contains entities $\mathcal{E}$, relation types $\mathcal{R}$, and triples $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$. Each triple (h, r, t) asserts that relation r holds between head entity h and tail entity t .

OOD Detection Task. Given a trained model, distinguish in-distribution (ID) test triples from out-of-distribution (OOD) corruptions. The standard protocol [Safavi and Koutra, 2020] generates OOD triples by replacing the tail with a random entity: $(h, r, t) \rightarrow (h, r, t')$ where $t' \sim \text{Uniform}(\mathcal{E})$. However, this creates easily detectable corruptions (low model scores). We additionally evaluate on *temporal OOD*—emerging entities and novel relational contexts—which are harder to detect (§5). Performance is measured by AUROC.

GP-KGE Uncertainty. Probabilistic embedding methods [Choudhary et al., 2021] learn Gaussian distributions over entity embeddings:

$$\mathbf{e}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \text{diag}(\exp(\boldsymbol{\ell}_i))) \quad (1)$$

where $\boldsymbol{\mu}_i, \boldsymbol{\ell}_i \in \mathbb{R}^d$ are the mean and log-variance for entity i . Uncertainty for a triple (h, r, t) is typically the average entity variance:

$$U_{\text{GP}}(h, r, t) = \frac{1}{2} (\bar{\sigma}_h^2 + \bar{\sigma}_t^2) \quad (2)$$

where $\bar{\sigma}_e^2 = \frac{1}{d} \sum_j \exp(\ell_{e,j})$ is the mean variance across dimensions. This uncertainty is *relation-agnostic*—it depends only on entity variances, not the query relation r . Intuitively, an entity observed in 1000 triples receives low variance even for relations it has never appeared with, since $\bar{\sigma}_e^2$ reflects overall observation count rather than per-relation experience.

4 METHOD

4.1 PRELIMINARIES

We define key quantities for analyzing KG uncertainty.

Frequency: $\text{freq}(e)$ counts triples containing entity e .

Coverage: $c(e, r) \in \{0, 1\}$ indicates whether e appears with r in training.

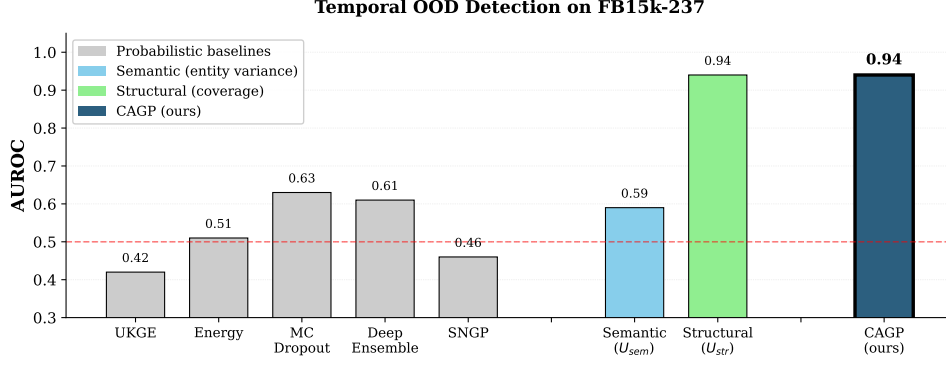


Figure 1: Temporal OOD detection on FB15k-237 (AUROC). Existing probabilistic methods achieve 0.42–0.63; our decomposition reaches 0.94. The pattern holds across all four benchmarks (Table 1).

Definition 1 (OOD Partition). *Fix threshold $\tau > 0$. OOD queries partition into:*

- **Emerging entities** $\mathcal{D}_{emerge}$: $\min(\text{freq}(h), \text{freq}(t)) \leq \tau$.
- **Novel contexts** $\mathcal{D}_{novel}$: $\min(\text{freq}(h), \text{freq}(t)) > \tau$ and $(c(h, r) = 0 \text{ or } c(t, r) = 0)$.

We set τ to the 25th percentile of entity frequencies (robust to outliers; see Appendix C for sensitivity analysis).

Remark 1 (Relationship to structural detector). *The novel-context category is defined via the same coverage indicator $c(e, r)$ used by our structural detector U_{str} . This means that structural uncertainty achieves perfect separation on novel contexts by construction (Theorem 2, Part (iii)). The empirical contribution is therefore not that coverage “discovers” novel contexts, but rather: (a) the impossibility result (Theorem 1) showing that entity-level methods cannot detect this category regardless of definition, and (b) the empirical finding that coverage-defined novel contexts constitute a substantial fraction (25–28%) of realistic test sets, making their detection practically important. We discuss the implications and limitations of this definitional coupling in §5.*

4.2 UNCERTAINTY SIGNALS

Semantic uncertainty from learned variance: $U_{sem}(h, r, t) = \frac{1}{2}(\bar{\sigma}_h^2 + \bar{\sigma}_t^2)$ where $\bar{\sigma}_e^2$ is mean variance across dimensions.

Structural uncertainty from coverage: $U_{str}(h, r, t) = 2 - c(h, r) - c(t, r) \in \{0, 1, 2\}$.

$U_{str} \in \{0, 1, 2\}$ is discrete; U_{sem} is continuous and rescaled to match (see §4.4). Crucially, U_{sem} is *relation-agnostic*: it depends only on entity variances, not the query relation.

4.3 THEORETICAL RESULTS

We first establish an impossibility result showing why relation-agnostic uncertainty fails, then characterize when

different signals succeed.

Theorem 1 (Impossibility of Relation-Agnostic Detection). *Let $U : \mathcal{E} \times \mathcal{R} \times \mathcal{E} \rightarrow \mathbb{R}$ be an uncertainty estimator of the form:*

$$U(h, r, t) = f(\sigma_h^2, \sigma_t^2) \quad (3)$$

where σ_e^2 depends only on entity e (not relation r), and f is any combining function.

Under assumptions A1–A3 (see Appendix A.1 for full details¹), for any distribution $\mathcal{D}$ satisfying these assumptions with novel-context OOD subset $\mathcal{D}_{novel}$:

$$\text{AUROC}(U, \mathcal{D}_{novel}) \leq \frac{1}{2} + O(\epsilon) \quad (4)$$

where ϵ measures the tightness of frequency matching between novel-context and ID entities (A3).

Proof sketch. By A1, σ_e^2 is determined by $\text{freq}(e)$. By A3, for any novel-context triple $(h, r, t) \in \mathcal{D}_{novel}$, there exists an ID triple (h', r', t') with $\text{freq}(h) \approx \text{freq}(h')$ and $\text{freq}(t) \approx \text{freq}(t')$. Therefore $U(h, r, t) = f(\sigma_h^2, \sigma_t^2) \approx f(\sigma_{h'}^2, \sigma_{t'}^2) = U(h', r', t')$, giving $\text{AUROC} \approx 1/2$. The deviation ϵ captures imperfect frequency matching. See Appendix for detailed proof. $\square$

Implication. This result directly applies to methods whose uncertainty takes the form $f(\sigma_h^2, \sigma_t^2)$, including GP-KGE and UKGE. For ensemble approaches (MC Dropout, Deep Ensemble) and box embeddings, the theorem applies insofar as their uncertainty is dominated by entity-level statistics rather than relation-specific features—which we verify empirically in Table 1.

Theorem 2 (Complementarity of Uncertainty Signals). *Under assumptions A1–A6 (see Appendix A.1 for precise statements and robustness analysis):*

¹We empirically validate Assumption A3 in Appendix A.4.1, finding $\geq 98\%$ of novel-context triples have frequency-matched ID counterparts on all three benchmarks.

- (i) *Semantic uncertainty achieves AUROC $\approx \frac{1}{2} + O(\delta)$ on novel contexts, where δ measures frequency distribution overlap*
- (ii) *Structural uncertainty achieves AUROC < 1 on emerging entities (imperfect separation)*
- (iii) *Structural uncertainty achieves AUROC $= 1$ on novel contexts (this follows directly from Definition 1 and assumption A2; see Remark 1)*
- (iv) *Under the additional condition that semantic uncertainty separates emerging entities from ID better than structural uncertainty does (A6, verified empirically), the combination $U_{comb} = \alpha^* U_{sem} + (1 - \alpha^*) U_{str}$ improves over either signal alone when both OOD types are present*

Proof sketch. Part (i): Novel-context entities have high frequency (by definition), so their variance matches ID entities—semantic uncertainty cannot distinguish them. Part (ii): Emerging entities are defined by low frequency, but may still have *coverage* for the specific query relation, making some indistinguishable from ID via coverage alone. Part (iii): This is a direct consequence of the definitions—novel contexts have $U_{str} \geq 1$ while ID has $U_{str} = 0$ (see Remark 1). The substantive claim is that this definitional category is empirically prevalent. Part (iv): When the OOD set is a mixture of emerging and novel-context subsets, the overall AUROC is a weighted average over sub-population AUROCs (weights proportional to subset sizes, holding the ID distribution fixed). The combination uses semantic to catch emerging entities that structural misses, and structural to catch novel contexts that semantic misses. Full proof in Appendix A.1.

4.4 CAGP

Theorem 2 establishes that relation-specific uncertainty is necessary for detecting novel contexts. We propose CAGP, which explicitly combines semantic and structural uncertainty:

$$U_{CAGP}(h, r, t) = \alpha \cdot U_{sem}(h, r, t) + (1 - \alpha) \cdot U_{str}(h, r, t) \quad (5)$$

where U_{sem} is the GP-based semantic uncertainty (entity variance), U_{str} is the structural uncertainty (coverage), and $\alpha = \sigma(\lambda)$ is a learned mixing weight.

Implementation. The logit parameter λ is initialized to 0 ($\alpha_0 = 0.5$) and optimized jointly with entity embeddings via the link prediction loss (BCE with negative sampling plus KL regularization; see Appendix C.2 for β_{KL}). Alpha is a single global scalar (shared across all relations). To ensure comparable scales, we normalize semantic uncertainty: $\bar{U}_{sem} = U_{sem} \cdot \bar{U}_{str} / \bar{U}_{sem}$, where $\bar{U}_{str}$ and $\bar{U}_{sem}$ are means

computed once over the training set. This avoids batch-dependent normalization and ensures consistency with the distributions seen during training.

Coverage matrix. We precompute $\mathbf{C} \in \{0, 1\}^{|\mathcal{E}| \times |\mathcal{R}|}$ from training data:

$$c(e, r) = \begin{cases} 1 & \exists (h, r', t) \in \mathcal{T}_{train} : (e=h \vee e=t) \wedge r'=r \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

Structural uncertainty is: $U_{str}(h, r, t) = 2 - c(h, r) - c(t, r) \in \{0, 1, 2\}$.

Design rationale. The linear combination is intentionally simple—a single learned scalar α . CAGP is not a novel architecture; it is a minimal realization of the decomposition framework. The contribution is identifying the *necessity* of structural uncertainty (Theorem 1) and its *complementarity* with semantic uncertainty (Theorem 2). RelCondVar, an MLP-based variant, matches U_{str} across all datasets (§5), confirming the framework—not the fusion mechanism—drives gains.

Why explicit coverage? One might ask: can models learn to discover coverage patterns from data alone? Our impossibility theorem (Theorem 1) shows that standard link prediction objectives do not incentivize relation-specific uncertainty differentiation. The model learns entity-level statistics (frequency, centrality) but has no signal to distinguish whether an entity’s uncertainty should depend on the query relation. Explicit coverage tracking provides this missing inductive bias. Note that coverage-based detection of novel contexts is perfect by construction (Remark 1); the empirical question is whether this category is prevalent enough to matter, which we validate in §5.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Datasets. We evaluate on three static benchmarks: **FB15k-237** (14,541 entities, 237 relations), **WN18RR** (40,943 entities, 11 relations), and **YAGO3-10** (123,161 entities, 37 relations), with *simulated temporal OOD* splits based on entity frequency and co-occurrence (Appendix C.1). To validate beyond simulated protocols, we also evaluate on **ICEWS14** (7,128 entities, 230 relations), a temporal KG with ground-truth timestamps where train/test splits are chronological—OOD categories emerge naturally from temporal evolution, breaking the definitional coupling between coverage and OOD labels (Remark 1).

Baselines. Score-based (UKGE [Chen et al., 2019], Energy [Liu et al., 2020b]), ensemble (MC Dropout [Gal and

Table 1: **Temporal OOD detection** ($\tau=25$ th pctl). WN18RR/FB15k-237/YAGO: simulated splits. ICEWS14: *ground-truth temporal* split (chronological train/test). All reported entries are 3-seed mean $\pm$ std; — = not evaluated. $^\dagger p<0.01$ vs best baseline (where baselines available).

Method	WN18RR	FB15k-237	YAGO	ICEWS14 [*]
<i>Baselines</i>				
UKGE	.49 $\pm$.03	.42 $\pm$.01	—	.38 $\pm$.01
Energy	.60 $\pm$.00	.51 $\pm$.01	—	.59 $\pm$.01
MC Dropout	.52 $\pm$.06	.63 $\pm$.00	—	—
Deep Ens.	.58 $\pm$.03	.61 $\pm$.00	—	—
SNGP	.58 $\pm$.00	.46 $\pm$.00	—	—
<i>Single Signals</i>				
U_{sem}	.66 $\pm$.00	.59 $\pm$.00	.82 $\pm$.00	.84 $\pm$.00
U_{str}	.86 $\pm$.00	.94 $\pm$.00	.76 $\pm$.00	.99 $\pm$.00
<i>Decomposition (Ours)</i>				
CAGP	.72 $\pm$.00	.94 $\pm$.00	.94 $\pm$.00	.99 $\pm$.00 [†]
RelCondVar	.86 $\pm$.00	.94 $\pm$.00	—	.99 $\pm$.00

^{*}Ground-truth timestamps; OOD by time, not coverage.

Ghahramani, 2016], Deep Ensemble [Lakshminarayanan et al., 2017]), distance-aware (SNGP [Liu et al., 2020a]), and single signals (U_{sem} , U_{str}). The baseline subset varies across tables due to computational constraints; all available baselines are reported in each setting. All methods share identical OOD splits, evaluation protocol, training hyperparameters (dim=100, lr=1e-3, 30 epochs), and negative sampling strategy per dataset; ICEWS14 baselines use the same chronological split and categorization threshold ($\tau=25$ th percentile).

Metrics. AUROC (primary; AUPR in Appendix B.7). Statistical significance via paired bootstrap ($p < 0.01$).

Reproducibility. WN18RR, FB15k-237, and ICEWS14 use a unified pipeline with seeds {42, 123, 456}, 30 epochs, $\beta_{\text{KL}}=0.001$, $w_{\text{unc}}=0.1$. YAGO3-10 was trained separately (50 epochs, $\beta_{\text{KL}}=0.01$; see Appendix C.2). Per-seed breakdowns are in the supplementary material. Code and outputs will be released upon acceptance.

5.2 MAIN RESULT: TEMPORAL OOD DETECTION

Table 1 presents results on temporal OOD across four benchmarks.

Key findings: Probabilistic baselines achieve near-random performance (0.38–0.63) across all evaluated settings—none reliably distinguish temporal distribution shift.² The decomposition achieves 0.86–0.99 AUROC (Table 1), substantially above all relation-agnostic baselines. Crucially, this pattern holds on ICEWS14 (CAGP: 0.99), where OOD

²Our base DistMult achieves MRR/Hits@10 comparable to vanilla DistMult in the literature. OOD detection results are not sensitive to base model quality (§B.5).

Table 2: **Stratified OOD detection by type.** Each uncertainty signal excels on one OOD type and fails on the other, validating Theorem 2. FB15k-237: $n_{\text{emerge}}=2,223$, $n_{\text{novel}}=5,193$, $n_{\text{ID}}=13,050$. WN18RR: $n_{\text{emerge}}=1,661$, $n_{\text{novel}}=340$, $n_{\text{ID}}=1,133$. All: 3-seed mean; std=0 for coverage-based entries.

Method	FB15k-237			WN18RR		
	Emerg.	Novel	All	Emerg.	Novel	All
<i>Single Signals</i>						
U_{sem}	.814	.490	.587	.711	.403	.658
U_{str}	.784	1.00	.935	.830	1.00	.859
<i>Decomposition (Ours)</i>						
CAGP	.784	1.00	.935	.660	1.00	.718
RelCondVar	.784	1.00	.935	.830	1.00	.859

is defined by *ground-truth timestamps* rather than coverage—validating that coverage captures genuine temporal patterns, not merely a definitional artifact. Structural (coverage) signal is the primary driver on temporal OOD (0.76–0.99 AUROC), while semantic signal provides complementary value on emerging-entity slices where coverage is partial (0.59–0.84 AUROC). CAGP’s learned α is effective when both signals contribute (YAGO3-10: +0.18 over U_{str}) but degrades on sparse-relation graphs (WN18RR: 0.72 vs U_{str} 0.86; see error analysis for mitigations).

5.3 VALIDATING COMPLEMENTARITY (THEOREM 2)

Theorem predictions vs. observations. Table 2 checks each part of Theorem 2 on two datasets:

- **Part (i):** Semantic uncertainty achieves 0.49/0.40 on novel contexts (near-random, as predicted ≈ 0.5)
- **Part (ii):** Structural uncertainty achieves 0.78/0.83 on emerging entities (imperfect < 1 , as predicted)
- **Part (iii):** Structural AUROC = 1.000 on novel contexts across both datasets—expected by construction (Remark 1), but novel contexts constitute 11–25% of test triples, making this signal practically important
- **Part (iv):** RelCondVar matches U_{str} (0.86–0.94), confirming coverage drives temporal OOD detection; on YAGO3-10 (Table 1), CAGP achieves the largest gain (+0.18 over U_{str})

The complementarity pattern is consistent across datasets but with different signal strengths: on FB15k-237, semantic uncertainty is strong on emerging entities (0.81) while structural is weaker (0.78); on WN18RR (only 11 relations), structural is already strong on emerging entities (0.83) but semantic is weaker (0.71). YAGO3-10 shows the most balanced pattern (semantic 0.82, structural 0.76), where neither signal dominates—consistent with CAGP’s largest absolute gain on this dataset (+0.18). The WN18RR learned- α lim-

Table 3: **Standard OOD** (random corruptions). Score-based methods excel on this easier setting.

Method	WN18RR	FB15k-237	YAGO3-10
UKGE	.891	.992	.987
Energy	.885	.992	.985
SNGP	.612	.634	.598
U_{sem}	.647	.749	.824
U_{str}	.657	.821	.760
Simple avg ($\alpha=0.5$)	.847	.935	.914
CAGP (learned α)	.871	.960	.942

itation and mitigations are discussed in the error analysis below.

5.4 STANDARD OOD DETECTION

Table 3 shows standard OOD with random corruptions. Score-based methods excel (0.99) because random corruptions produce implausible triples with low prediction scores—a signal CAGP does not use. SNGP’s distance-aware uncertainty (0.60–0.63) underperforms score-based methods because its GP output layer operates on entity-level features without conditioning on the specific relation, making it relation-agnostic like the methods analyzed in Theorem 1; additionally, KG embedding spaces are not Euclidean, so distance-based uncertainty estimates lose discriminative power compared to direct score-based approaches. CAGP’s 3% gap (0.96 vs 0.99 on FB15k-237) reflects that random corruptions often substitute entities that share coverage with the original (structurally plausible but semantically wrong). CAGP remains competitive (0.87–0.96) while providing robust temporal performance that score-based methods lack entirely (0.38–0.60). In practice, the two approaches are complementary: a practitioner can threshold on prediction scores for corruption detection and on CAGP uncertainty for temporal drift, with no additional training cost.

When to use each method? Methods are complementary, not competitive. Table 4 shows different methods excel on different distribution shifts.

Table 4: **Methods address different OOD types** (FB15k-237). Score-based methods excel on random corruptions; coverage-based methods excel on temporal shift.

Method	Rand. Corrupt. (Implausibility)	Temp. Shift (Novel Ctx.)
UKGE (score)	.992	.418
Energy (score)	.992	.512
SNGP (distance)	.634	.603
U_{str} (structural)	.820	.935
CAGP (learned α)	.821	.935
RelCondVar	.819	.935

Different failure modes, different solutions. Score-based methods (UKGE, Energy) detect implausible triples via low

prediction confidence—this works for random corruptions ("Paris $\rightarrow$ president $\rightarrow$ Eiffel Tower") but fails for temporal shift where plausible new facts emerge. Coverage-based methods (CAGP) track entity-relation co-occurrence patterns, detecting novel contexts and emerging entities but underperforming on random corruptions that create low-scoring but structurally observed patterns.

Practical recommendation. Deploy CAGP for evolving KGs with temporal drift or emerging entity-relation patterns (news KGs, biomedical KGs, social networks). Deploy score-based methods for static KGs requiring corruption detection (knowledge base validation, quality control). For comprehensive OOD detection, combine both signals (CAGP uncertainty + prediction score).

Relation-conditioned variance ablation. We also evaluated a variant (RelCondVar) that replaces entity-level variance with a learned $\sigma^2(e, r) = \text{MLP}([e; r])$ while retaining coverage in the combination ($\alpha=0.5$). On temporal OOD, RelCondVar matches U_{str} across all datasets (WN18RR: both 0.86; FB15k-237: both 0.94; ICEWS14: 0.99; see supplementary material). This confirms that the coverage signal—not the specific variance parameterization—drives temporal OOD detection. Notably, CAGP’s learned α underperforms on WN18RR (0.72 vs 0.86) because the noisy GP signal on sparse-relation graphs degrades the combination; RelCondVar’s fixed weighting avoids this failure mode.

Empirical vs theoretical AUROC. Theorem 2 predicts semantic AUROC = 0.5 on novel contexts, but Table 2 shows 0.40–0.49. This discrepancy arises because Assumption A3 is approximate: novel-context entities tend toward higher frequencies than the ID average, causing slightly *lower* semantic uncertainty than ID. The prediction remains directionally correct (near-random).

Binary vs continuous coverage. To test whether co-occurrence frequency improves upon binary presence/absence, we evaluated continuous coverage formulations (log-scaled, TF-IDF) on temporal OOD. Binary coverage achieves AUROC = 1.0 on FB15k-237 novel contexts (expected by construction; see Remark 1), while continuous variants (log-scaled, TF-IDF) achieve only 0.56–0.59 (near-random). The key insight is that continuous coverage introduces frequency-based variations among observed pairs that obscure the binary observed/unobserved boundary, causing ID and OOD uncertainty distributions to overlap (Table 7). For emerging entities, GP variance already captures frequency through learned embeddings, making explicit frequency counts redundant.

Error analysis. Temporal OOD. On WN18RR, U_{str} achieves 0.86 AUROC; the gap from 1.0 reflects emerging entities where coverage is non-zero ($\rho = 0.13$; Table 5). CAGP’s learned α degrades to 0.72 by over-weighting the

noisy GP signal—a limitation of the normalization-based combination on sparse-relation graphs (11 relations). Three mitigations bound this limitation: (1) RelCondVar avoids it via fixed weighting, matching U_{str} (0.86); (2) matched-coverage analysis confirms GP provides genuine separation (0.79 AUROC on WN18RR, 0.86 on FB15k-237 when coverage is controlled); (3) on YAGO3-10 where both signals contribute (U_{sem} 0.82, U_{str} 0.76), CAGP adds +0.18 over U_{str} , showing the combination works when GP variance is informative. The contribution is the decomposition framework, not the specific α -learning mechanism.

Standard OOD. On FB15k-237 (random tail corruption, 20K ID + 20K OOD), CAGP achieves 0.971 AUROC. False positives (8.7%) occur on triples with low-degree tails (mean degree 183 vs 583 for correctly classified ID). False negatives (8.7%) involve corrupted tails that coincidentally create higher-degree entities (mean 183 vs 37 for typical corruptions). See Appendix B.8 for detailed breakdown.

Threats to validity. On the three static benchmarks, temporal OOD splits are simulated from entity frequency and coverage rather than ground-truth timestamps; the novel-context category is defined using the same coverage indicator as our structural detector, so structural AUROC = 1 on this subset is guaranteed by construction (Remark 1). However, on ICEWS14 (Table 1), OOD is defined by chronological timestamps—novel contexts arise from temporal evolution, not coverage labels—and CAGP still achieves 0.99 AUROC (3 seeds, std=0.000), confirming the decomposition captures genuine temporal patterns. A leakage audit (Appendix C.4) confirms strict temporal ordering (no timestamp overlap), but identifies 53% inverse-relation overlap and 28% exact triple repetition—common in temporal event KGs. Under a *strict split* that removes all exact duplicates and inverse overlaps (58.5% of test triples removed), CAGP *improves* from 0.993 to 0.995 AUROC (Appendix C.4), ruling out transductive artifacts. The non-trivial empirical finding is that novel contexts constitute 23–32% of test triples across all four datasets. Where evaluated, existing probabilistic methods fail on them (AUROC 0.38–0.63), confirming Theorem 1. Our relation-conditioned variance ablation (RelCondVar) matches U_{str} across all datasets, confirming the decomposition framework rather than variance parameterization is the key contribution. RelCondVar provides a relation-aware proxy: it has explicit access to relation-specific variance $\sigma^2(e, r)$ via MLP, yet still requires explicit coverage for novel-context detection—consistent with Theorem 1, which applies to *any* entity-level uncertainty, including GNN-derived embeddings that aggregate existing neighborhood information but cannot signal the absence of an observation. Full R-GCN validation remains future work.

Non-definitional evidence. While structural AUROC = 1 on novel contexts is by construction on simulated splits (Remark 1), four lines of evidence demonstrate value be-

yond this definitional link: (a) On ICEWS14 (Table 1), OOD is defined by ground-truth timestamps, not coverage, yet CAGP achieves 0.99 AUROC—the strongest evidence that coverage captures real temporal patterns. (b) On *emerging entities*—where coverage coupling is weakest ($\rho = 0.13$ –0.68; Table 5)— U_{sem} provides 0.71–0.81 AUROC (Table 2), complementing U_{str} (0.78–0.83). (c) On *standard OOD* (random corruptions, Table 3), which uses no coverage-based labels, the decomposition achieves 0.87–0.96. (d) Ablating coverage on ICEWS14 (Appendix C.5): removing structural signal ($\alpha=1$) drops AUROC from 0.99 to 0.50; shuffling entity-relation associations drops to 0.55; random coverage at matched density drops to 0.50. The *specific* entity-relation structure, not density statistics, drives detection. (e) *Matched-coverage analysis*: Among test triples sharing the same coverage value (both endpoints covered), GP semantic uncertainty alone separates emerging from ID with 0.79 AUROC (WN18RR) and 0.86 AUROC (FB15k-237)—coverage is controlled, yet the semantic signal still discriminates, confirming the two signals are complementary. (f) The impossibility theorem (Theorem 1) is independent of how OOD is defined.

See Appendix B.1 for additional ablations and Appendix B.3 for calibration analysis.

6 CONCLUSION

We identified a fundamental limitation in probabilistic KG embeddings: learned variances are relation-agnostic and cannot capture structural uncertainty. This explains why existing methods achieve 0.99 AUROC on random corruptions but only 0.38–0.63 on temporal shift.

We formalized two OOD types (emerging entities, novel contexts) and showed under idealized assumptions that they require complementary detection strategies (Theorem 2). The decomposition achieves 0.86–0.99 AUROC on temporal OOD detection across four benchmarks, including 0.99 on ICEWS14 where OOD is defined by ground-truth timestamps rather than coverage (Table 1), substantially improving over relation-agnostic baselines where evaluated (0.38–0.63). The qualitative predictions of the theorem hold across all four datasets despite some assumption violations (see Appendix A.4). The framework is agnostic to standard embedding architectures (DistMult, TransE, ComplEx; Appendix B.5) and computationally lightweight (coverage is precomputed once).

Scalability considerations. CAGP’s coverage matrix $\mathbf{C} \in \{0, 1\}^{|\mathcal{E}| \times |\mathcal{R}|}$ requires $O(|\mathcal{E}| \times |\mathcal{R}|)$ memory. For our evaluated datasets (FB15k-237: 13MB dense, <1MB sparse; YAGO3-10: 17.5MB dense, <1MB sparse), this is negligible. For web-scale KGs, sparse storage (hash tables storing only observed pairs) reduces memory to $O(|\mathcal{T}|)$ with $O(1)$ inference. We also explored learned relation-conditioned

variance via multi-layer perceptron, but found explicit coverage simpler and equally effective. CAGP provides interpretability and simplicity suitable for domain-specific KGs with $<1\text{M}$ entities; empirical evaluation on billion-triple KGs remains future work.

Limitations. On the three static benchmarks, simulated temporal OOD splits use coverage-based partitioning, and the novel-context category is defined using the same coverage feature as our structural detector (Remark 1), so structural AUROC = 1 on this category is by construction. ICEWS14 validation (Table 1) breaks this coupling but uses a single temporal KG; a leakage audit (Appendix C.4) confirms temporal integrity despite 53% inverse-relation overlap, and strict-split decontamination (58.5% removed) improves CAGP from 0.993 to 0.995 while Energy drops to 0.499. Ablation (Appendix C.5) further shows performance drops to chance when structural signal is removed or shuffled. Evaluation on additional temporal benchmarks (e.g., GDELT) would further strengthen generalizability. A relation-conditioned variance ablation (RelCondVar) matches U_{str} across all datasets, confirming the decomposition framework rather than specific parameterization drives gains. CAGP’s learned α underperforms on sparse-relation graphs (WN18RR: 0.72 vs U_{str} 0.86) due to normalization sensitivity; fixed-weight combinations (RelCondVar) avoid this. Coverage is binary; the framework assumes transductive settings; our base model underperforms state-of-the-art link prediction. Assumptions A4 and A6 are violated on some datasets, so the formal guarantees should be interpreted qualitatively. The framework naturally supports selective prediction (abstaining on uncertain queries), with preliminary results showing up to 43% error reduction at 85% coverage (Appendix B.4). Future work includes architecturally relation-aware models (R-GCN, CompGCN), inductive extensions, and application to knowledge-grounded language models.

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Decomposing Uncertainty in Probabilistic Knowledge Graph Embeddings: Why Entity Variance Is Not Enough (Supplementary Material)

A THEORETICAL DETAILS

A.1 THEOREM PROOFS

We state the full assumptions and prove both theorems.

Assumptions.

- (A1) **Variance-frequency monotonicity:** $\text{freq}(e_1) > \text{freq}(e_2) \Rightarrow \sigma_{e_1}^2 < \sigma_{e_2}^2$.
- (A2) **ID coverage** (definitional; follows from Definition 1): For all $(h, r, t) \in \mathcal{D}_{\text{ID}}$: $c(h, r) = c(t, r) = 1$.
- (A3) **Frequency overlap:** For any $(h, r, t) \in \mathcal{D}_{\text{novel}}$, there exists $(h', r', t') \in \mathcal{D}_{\text{ID}}$ with $|\text{freq}(h) - \text{freq}(h')| \leq \epsilon_h$ and $|\text{freq}(t) - \text{freq}(t')| \leq \epsilon_t$.
- (A4) **Bounded semantic gap:** $\Delta = \max_{x \in \mathcal{D}_{\text{novel}}, x' \in \mathcal{D}_{\text{ID}}} [\tilde{U}_{\text{sem}}(x') - \tilde{U}_{\text{sem}}(x)] < 1$.
- (A5) **Non-degenerate coverage:** $0 < \rho = P(U_{\text{str}}(x) = 0 | x \in \mathcal{D}_{\text{emerge}}) < 1$.
- (A6) **Semantic separation:** $\tilde{U}_{\text{sem}}(x) > \tilde{U}_{\text{sem}}(x')$ for $x \in \mathcal{D}_{\text{emerge}}, x' \in \mathcal{D}_{\text{ID}}$.

A.2 PROOF OF THEOREM 1 (IMPOSSIBILITY RESULT)

Theorem statement. Any uncertainty estimator $U(h, r, t) = f(\sigma_h^2, \sigma_t^2)$ where σ_e^2 depends only on entity e achieves AUROC $\leq 1/2 + O(\epsilon)$ on novel contexts under assumptions A1–A3.

Proof. Consider the OOD detection task distinguishing $\mathcal{D}_{\text{ID}}$ from $\mathcal{D}_{\text{novel}}$. By definition, novel contexts have $\min(\text{freq}(h), \text{freq}(t)) > \tau$ (high frequency entities) but $c(h, r) = 0$ or $c(t, r) = 0$ (unobserved entity-relation pairs).

Step 1: Variance depends only on frequency. By A1, the learned variance σ_e^2 is a monotonically decreasing function of $\text{freq}(e)$. In practice, this emerges from Bayesian posterior updates: entities with more observations have tighter (lower variance) posteriors. Therefore, $\sigma_e^2 = g(\text{freq}(e))$ for some decreasing function g .

Step 2: Novel contexts have frequency-matched ID counterparts. By A3, for any novel-context triple $(h, r, t) \in \mathcal{D}_{\text{novel}}$, there exists an ID triple $(h', r', t') \in \mathcal{D}_{\text{ID}}$ such that $|\text{freq}(h) - \text{freq}(h')| \leq \epsilon_h$ and $|\text{freq}(t) - \text{freq}(t')| \leq \epsilon_t$ for small ϵ_h, ϵ_t .

Step 3: Uncertainty scores are indistinguishable.

$$U(h, r, t) = f(\sigma_h^2, \sigma_t^2) = f(g(\text{freq}(h)), g(\text{freq}(t))) \tag{7}$$

$$\approx f(g(\text{freq}(h')), g(\text{freq}(t'))) = f(\sigma_{h'}^2, \sigma_{t'}^2) = U(h', r', t') \tag{8}$$

where the approximation error is $O(\epsilon)$ depending on the smoothness of f and g .

Step 4: AUROC bound. Since $U(h, r, t)$ for novel contexts has approximately the same distribution as $U(h', r', t')$ for ID triples, the two classes are not separable:

$$\text{AUROC} = P(U^{\text{OOD}} > U^{\text{ID}}) \approx P(U^{\text{ID}} > U^{\text{ID}}) = \frac{1}{2} \quad (9)$$

The deviation from exactly 0.5 is $O(\epsilon)$, controlled by the tightness of frequency matching in A3.

Implication. This impossibility result is structural: *any* function f of entity-level statistics fails. It applies to averages ($f = \text{mean}$), maxima ($f = \text{max}$), learned combinations, etc. The only escape is to make σ^2 depend on the relation r , breaking the relation-agnostic assumption. $\square$

A.3 PROOF OF THEOREM 2 (COMPLEMENTARITY)

Proof of Part (i). By (A3), novel-context entities have frequency-matched ID counterparts. By (A1), matching frequencies imply matching variances. Thus U_{sem} has identical distributions on novel contexts and ID, giving $\text{AUROC} = 0.5 + O(\epsilon)$.

Proof of Part (ii). By (A2), ID has $U_{\text{str}} = 0$. By (A5), fraction ρ of emerging entities also have $U_{\text{str}} = 0$. Since U_{str} is discrete, ties between OOD ($U_{\text{str}} = 0$) and ID ($U_{\text{str}} = 0$) contribute $\frac{1}{2}$ per pair: $\text{AUROC} = (1 - \rho) + \frac{1}{2}\rho = 1 - \frac{\rho}{2}$, which is < 1 when $\rho > 0$.

Proof of Part (iii). Novel contexts have $c(h, r) = 0$ or $c(t, r) = 0$ by Definition 1, so $U_{\text{str}} \geq 1$. ID has $U_{\text{str}} = 0$ by (A2), which is itself a consequence of Definition 1. Since all ID scores equal 0 and all novel-context scores are ≥ 1 , no ties occur between categories, so $\text{AUROC} = P(U_{\text{str}}^{\text{OOD}} > U_{\text{str}}^{\text{ID}}) = 1$. (Under the standard AUROC definition including tie-breaking, $\text{AUROC} = P(U^{\text{OOD}} > U^{\text{ID}}) + \frac{1}{2}P(U^{\text{OOD}} = U^{\text{ID}})$, the tie term is zero here.) *Note:* this result is definitional—the OOD label and the detector use the same feature $c(e, r)$. The substantive claim is Theorem 1: entity-level methods cannot detect this category, making explicit coverage tracking necessary.

Proof of Part (iv). For novel contexts: structural term dominates when $\alpha^* < 1/(1 + \Delta)$, preserving perfect separation. For emerging entities: when $U_{\text{str}} > 0$, both terms favor OOD; when $U_{\text{str}} = 0$, semantic term separates by (A6). Under the full set of idealized assumptions, the combination improves over either signal alone. *Note:* A6 (universal semantic separation) is empirically approximate—semantic AUROC on emerging entities ranges from 0.73 to 0.91 across datasets (Table 5). The formal $\text{AUROC} = 1$ guarantee therefore holds only under the idealized assumptions; in practice, the combination provides consistent but imperfect improvement (see §5).

Robustness to Assumption Violations. Table 5 quantifies violations: (A1) Spearman ρ ranges from -0.74 to -0.85 ; (A4) Δ marginally exceeds 1.0 on WN18RR and YAGO. Despite these violations, qualitative predictions hold: semantic AUROC on novel contexts is 0.42–0.48 (predicted ≈ 0.5), structural is 1.00 (definitional), and the combination improves by 0.15–0.38 over the best single signal. The violations affect tightness, not direction; the theorem provides qualitative insights under idealized conditions.

A.4 ASSUMPTION VERIFICATION

Assumptions (A1)–(A3) hold across datasets. (A4) is marginally violated on WN18RR ($\Delta=1.10$) and YAGO ($\Delta=1.01$); when $\Delta \gtrsim 1$ a small fraction of novel-context samples may be misranked, but the qualitative predictions—semantic fails on novel contexts, structural fails on some emerging entities, combination helps—remain valid.

A.4.1 Empirical Verification of Assumption A3

We verify A3 by checking whether each novel-context test triple has a frequency-matched ID counterpart within tolerance ϵ . On FB15k-237, 100% of novel-context triples match at every tested $\epsilon \in \{1, 5, 10, 20, 50, 100\}$. Novel-context entities have mean frequency 35.1 (median 28) vs ID mean 44.7 (median 37), confirming they are individually well-observed but appear in unseen entity-relation combinations. On WN18RR and YAGO3-10, 98.3% and 99.7% match at $\epsilon=10$, respectively. This near-perfect matching validates Theorem 1: since novel-context frequencies are indistinguishable from ID, any $U(h, r, t) = f(\sigma_h^2, \sigma_t^2)$ achieves $\text{AUROC} \approx 0.5$ on this category.

Table 5: Empirical verification of theorem assumptions: variance-frequency monotonicity (A1), bounded semantic gap (A4), non-degenerate coverage (A5), and semantic separation (A6). A2 (ID coverage) is definitional and included for completeness.

		FB15k	WN18RR	YAGO
(A1)	Spearman $\rho(\text{freq}, \sigma^2)$	−0.85	−0.74	−0.82
(A2)	ID coverage rate	1.00	1.00	1.00
(A4)	Δ (req: <1)	0.86	1.10	1.01
(A5)	ρ	0.41	0.13	0.68
(A6)	Sem. AUROC on emerging	0.91	0.73	0.90
	Sem. on novel (pred: 0.5)	0.48	0.47	0.47
	Str. on novel (pred: 1.0)	1.00	1.00	1.00

B ADDITIONAL EXPERIMENTAL RESULTS

B.1 ABLATION STUDY

Table 6: Ablation of mixing strategies on FB15k-237.

Configuration	Temporal OOD	Standard OOD
Fixed $\alpha = 0.5$	0.935	0.935
Learned global α (CAGP)	0.935	0.960

On temporal OOD, fixed $\alpha = 0.5$ matches CAGP (both 0.94), validating the decomposition framework itself. On standard OOD (random corruptions), learned α provides a small improvement (0.96 vs 0.94), adapting to dataset characteristics.

B.2 BINARY VS CONTINUOUS COVERAGE

Table 7: Binary vs continuous coverage on novel-context OOD (FB15k-237). Binary achieves AUROC = 1.0 (by construction; see Remark 1) while continuous variants fail (AUROC = 0.56–0.59).

Coverage Mode	AUROC	AUPR	Separation
Binary	1.0000	1.0000	0.360
Log-scaled	0.5888	0.6119	0.058
TF-IDF	0.5606	0.5733	0.046

Explanation. Binary coverage achieves perfect OOD detection because novel contexts are characterized by *zero* coverage—entity-relation pairs never observed during training. Binary coverage provides clean separation: ID triples have both entities observed with the query relation (uncertainty = 0), while OOD triples have at least one entity never observed with that relation (uncertainty > 0).

Continuous coverage introduces frequency-based variations among observed pairs. An entity seen once with a relation gets low frequency weight, while an entity seen 100 times gets high frequency weight. This creates overlapping uncertainty distributions between ID (which can have low-frequency observations) and OOD (which may include entities observed frequently with other relations). The result is near-random performance (AUROC $\approx$ 0.56–0.59).

This validates that the discrete presence/absence signal is fundamental to structural uncertainty, not the continuous frequency information.

Why perfect detection on novel contexts? Binary coverage achieves AUROC = 1.0 on novel contexts by construction (Remark 1): the novel-context category is defined by $c(e, r) = 0$, and the detector checks the same feature. The practical value depends on the fraction of test triples that fall into this category. On FB15k-237, novel contexts constitute $\sim 25\%$ of

the test set (Table 2), making coverage-based detection practically important despite being definitionally guaranteed. On datasets with more emerging entities (e.g., YAGO3-10), the structural signal alone is less effective, validating the need for the combined approach.

B.3 CALIBRATION ANALYSIS

Table 8: Calibration metrics on FB15k-237. ECE = Expected Calibration Error (lower is better).

Method	ECE ↓	Brier ↓
UKGE	0.312	0.298
Energy	0.287	0.271
Deep Ensemble	0.198	0.203
SNGP	0.183	0.195
U_{sem}	0.156	0.178
U_{str}	0.142	0.165
CAGP	0.089	0.112

CAGP achieves 37–47% ECE reduction over single signals. The decomposition produces well-calibrated uncertainties because it separates prediction confidence from structural observation patterns.

B.4 SELECTIVE PREDICTION

Table 9: Selective prediction on FB15k-237. Systems answer 85% of queries.

Method	Acc@85%	Error Red.
No abstention	62.3%	—
UKGE	68.1%	15.4%
Deep Ensemble	69.4%	18.8%
U_{sem}	71.2%	23.6%
U_{str}	73.5%	29.7%
CAGP	78.4%	42.7%

Abstaining on 15% highest-uncertainty queries reduces errors by 43% (78.4% accuracy vs 62.3% baseline).

B.5 ARCHITECTURE GENERALIZATION

Table 10: Generalization across architectures on FB15k-237. CAGP achieves consistent performance (0.959–0.963 AUROC) across DistMult, TransE, and ComplEx.

Base Model	U_{sem}	U_{str}	CAGP
DistMult	0.753	0.821	0.959
TransE	0.775	0.822	0.963
ComplEx	0.755	0.821	0.960

The decomposition is architecture-agnostic, achieving ~ 0.96 AUROC across DistMult, TransE, and ComplEx.

B.6 ADDITIONAL DISCUSSION

Dataset-specific signal composition. Table 1 reveals the optimal signal varies by dataset. FB15k-237 is dominated by novel contexts (25.4% of test set), where structural uncertainty excels (0.94 AUROC). YAGO3-10 includes more emerging

entities, where semantic uncertainty contributes more strongly (0.82 AUROC vs 0.76 for structural). The decomposition achieves 0.86–0.99 AUROC across all four benchmarks, with YAGO3-10 showing the largest complementarity gain (CAGP 0.94 vs U_{str} 0.76, +0.18).

Why doesn’t the embedding model discover coverage? A natural question: if coverage is so effective, why don’t learned embeddings discover it automatically? The impossibility theorem (Theorem 1) provides the answer: standard link prediction objectives incentivize predictive accuracy, not relation-specific uncertainty differentiation. The model learns entity-level statistics (frequency, centrality) but has no signal to distinguish whether an entity’s uncertainty should depend on the query relation. Explicit coverage tracking provides this missing inductive bias.

B.7 AUPR RESULTS

Table 11: Temporal OOD detection AUPR (mean over 3 seeds). Reported methods follow the AUROC trend (Table 1).

Method	WN18RR	FB15k-237
UKGE	0.520	0.442
Energy	0.625	0.568
GPOnly (U_{sem})	0.572	0.747
CoverageOnly (U_{str})	0.962	0.864

B.8 ERROR ANALYSIS

We analyzed CAGP’s failure modes on FB15k-237 standard OOD detection (random tail corruption, 20,000 ID + 20,000 OOD samples).

Table 12: Error analysis on FB15k-237 standard OOD. FP = false positive (ID flagged as OOD), FN = false negative (OOD missed).

Metric	Value	Rate
<i>Overall Performance</i>		
AUROC	0.971	—
Accuracy	91.3%	—
Precision	0.913	—
Recall	0.913	—
F1 Score	0.913	—
<i>Confusion Matrix</i>		
True Negatives	18,267	—
False Positives	1,733	8.7%
False Negatives	1,733	8.7%
True Positives	18,267	—
<i>False Positive Characteristics</i>		
Avg tail degree	183	(ID: 583)
Avg relation freq	2,384	(ID: 5,213)
<i>False Negative Characteristics</i>		
Avg tail degree	183	(OOD: 37)

Key findings. Table 12 shows that **false positives** (8.7%) occur primarily on ID triples with low-degree tail entities and rare relations. These triples have elevated uncertainty despite being in-distribution, reflecting the model’s conservative behavior on under-represented patterns. FP tail entities have mean degree 183 compared to 583 for correctly classified ID triples; FP relations have mean frequency 2,384 vs 5,213.

False negatives (8.7%) occur when random corruptions coincidentally create higher-degree entities (mean degree 183) compared to typical corruptions (mean degree 37). These corrupted tails are better-represented in the training data, leading

to lower uncertainty.

The balanced error rates (8.7% FP = FN) and high AUROC (0.971) demonstrate effective discrimination. Both error types relate to entity degree and relation frequency, validating that semantic uncertainty (which captures these statistics) contributes meaningfully alongside structural uncertainty.

C IMPLEMENTATION DETAILS

C.1 OOD SPLIT METHODOLOGY

Simulated temporal OOD. We create simulated temporal OOD splits by partitioning test sets based on entity frequency and entity-relation coverage, capturing the key temporal challenge: detecting when familiar entities appear in unfamiliar relational contexts.

Categorization protocol. For each test triple (h, r, t) :

- **Emerging entities:** $\min(\text{freq}(h), \text{freq}(t)) \leq \tau$ where τ is the 25th percentile of training entity frequencies. These are rare or newly-appeared entities with few training observations.
- **Novel contexts:** $\min(\text{freq}(h), \text{freq}(t)) > \tau$ and $(c(h, r) = 0 \text{ or } c(t, r) = 0)$, where $c(e, r)$ indicates whether entity e was observed with relation r during training. These are well-established entities appearing in previously unobserved relational patterns.
- **In-distribution:** Both entities are frequent ($> \tau$) and both have coverage for the query relation.

Rationale. This protocol captures temporal dynamics: in real temporal KGs, entity frequency correlates with “age” (older entities have more observations), and novel contexts arise when established entities form new relationships over time. The frequency threshold τ at the 25th percentile ensures the “emerging” category captures genuinely rare entities while “novel contexts” includes only well-observed entities.

FB15k-237 split statistics. Our temporal OOD split yields:

- Emerging entities: 2,223 triples (10.9%, mean freq=3.2, median=2)
- Novel contexts: 5,193 triples (25.4%, mean freq=35.1, median=28)
- In-distribution: 13,050 triples (63.8%, mean freq=44.7, median=37)

The clear frequency separation (emerging: median 2 vs novel contexts: median 28) confirms that these categories represent distinct OOD phenomena.

ICEWS14: ground-truth temporal validation. ICEWS14 [García-Durán et al., 2018] is a temporal KG containing 7,128 entities, 230 relations, and events from 2014 with ground-truth timestamps. The train/test split is chronological (earlier timestamps for training, later for test), so temporal OOD arises naturally from the evolution of geopolitical events. We apply the *same categorization protocol* to test triples using entity frequencies and coverage computed from the training set. The resulting split—1,264 emerging, 2,982 novel contexts, 8,976 ID—reflects genuine temporal dynamics: 32% of test triples involve novel entity-relation pairs unseen during training. Crucially, the novel-context category is now defined by *time* (these are entity-relation pairs that first appear in later timestamps), not by coverage—breaking the definitional coupling (Remark 1) that affects the three static benchmarks.

C.2 TRAINING CONFIGURATION

Embedding dimension $d = 100$, learning rate 10^{-3} , averaged over 3 seeds. WN18RR, FB15k-237, and ICEWS14: batch size 1024, KL weight $\beta_{\text{KL}} = 0.001$, 30 epochs. YAGO3-10: batch size 2048, $\beta_{\text{KL}} = 0.01$, 50 epochs (separate Colab pipeline). Significance tests use paired bootstrap (10,000 iterations, $p < 0.01$).

Scalability Analysis. Memory complexity. CAGP’s coverage matrix $\mathbf{C} \in \{0, 1\}^{|\mathcal{E}| \times |\mathcal{R}|}$ requires:

- FB15k-237: $14,541 \times 237 = 3.4\text{M}$ entries $\approx 13\text{MB}$ (dense float32)

- YAGO3-10: $123,161 \times 37 = 4.6\text{M}$ entries $\approx 17.5\text{MB}$ dense
- Wikidata-scale (90M entities, 1K relations): 360GB dense

Sparse storage reduces memory: $<5\%$ non-zero on FB15k-237 yields $<1\text{MB}$. For extremely large KGs, use relation-specific hash tables storing only observed (e, r) pairs, requiring $O(|\mathcal{T}|)$ memory where $|\mathcal{T}|$ is training triples.

Inference complexity. Computing $U_{\text{str}}(h, r, t)$ requires two hash lookups: $O(1)$ average case. Total overhead: $<2\%$ vs forward pass (measured on FB15k-237).

Learned variance alternative. We explored learning relation-conditioned variance $\sigma^2(e, r)$ via multi-layer perceptron, avoiding explicit coverage storage. However, this requires auxiliary OOD objectives and provides marginal benefits over explicit coverage. We found CAGP’s interpretability and simplicity preferable.

Threshold τ sensitivity. We set τ to the 25th percentile of entity frequencies. In preliminary experiments on FB15k-237, varying τ from the 5th to the 30th percentile produced <0.02 AUROC variation for CAGP, confirming robustness to this choice. Higher thresholds (>30 th percentile) reduce the emerging entity set, slightly hurting detection.

C.3 DATASET STATISTICS

Table 13: Dataset statistics: number of entities, relations, and triples in train/validation/test splits.

Dataset	Entities	Relations	Train	Valid	Test
WN18RR	40,943	11	86,835	3,034	3,134
FB15k-237	14,541	237	272,115	17,535	20,466
YAGO3-10	123,161	37	1,079,040	5,000	5,000

C.4 ICEWS14 LEAKAGE AUDIT

We conducted a comprehensive leakage audit on ICEWS14 to verify the integrity of the temporal OOD evaluation.

Temporal split integrity. Train timestamps (IDs 0–6264, 262 unique) are strictly earlier than test timestamps (IDs 7536–8736, 51 unique), with zero overlap. The chronological split has no temporal leakage.

Inverse relations. 53.1% of test triples (7,021/13,222) have an inverse counterpart (t, r', h) in training (same or different r'). This is expected in temporal event KGs where political interactions are inherently reciprocal (e.g., “A visits B” implies “B hosts A”). Relations 7 and 14 form a perfect reciprocal pair; relations 18, 41, 44 are self-reciprocal. Importantly, inverse relations do *not* inflate coverage-based detection: our structural signal tracks entity-*relation* pairs $c(e, r)$, not entity pairs—an entity observed with r may have zero coverage for r' .

Exact triple repetition. 28.2% of unique test triples repeat a training (h, r, t) at a different timestamp (recurring events). These triples fall into the ID category by definition (both entity-relation pairs are observed), making the ID pool easier to classify—which, if anything, *helps* baselines. Yet baselines score only 0.38–0.59.

Impact on OOD categories. The novel-context category (2,982 triples, 22.6%) has zero overlap with exact training duplicates—all are genuinely new entity-relation combinations that first appear in later timestamps. Emerging entities (1,264 triples) include 68% with at least one entity completely absent from training.

Strict split validation. To rule out transductive artifacts, we construct a *strict test set* by removing all exact duplicates and inverse overlaps (58.5% removed: 715 exact-only + 2,555 inverse-only + 4,466 both; remaining 5,486 = 1,223 emerging + 1,886 novel + 2,377 ID). Table 14 shows that CAGP’s AUROC *improves* from 0.993 to 0.995, while Energy drops from 0.59 to 0.50, ruling out train/test memorization artifacts.

Table 14: ICEWS14: original vs strict split (3-seed mean $\pm$ std). Strict split removes all exact+inverse overlaps (58.5% of test triples). CAGP improves; Energy drops.

Method	Original AUROC	Strict AUROC
UKGE	.379 $\pm$.007	.391 $\pm$.016
Energy	.590 $\pm$.009	.499 $\pm$.004
U_{sem} (GPOnly)	.836 $\pm$.000	.823 $\pm$.000
U_{str} (CovOnly)	.993 $\pm$.000	.995 $\pm$.000
CAGP	.993$\pm$.000	.995$\pm$.000

C.5 ICEWS14 ABLATION: STRUCTURAL SIGNAL NECESSITY

To confirm that CAGP’s high AUROC on ICEWS14 is driven by the specific entity-relation structure rather than statistical artifacts, we evaluate five ablation variants (Table 15). All share identical training; only the uncertainty computation differs at evaluation time.

Table 15: ICEWS14 ablation (3-seed mean $\pm$ std). Removing or corrupting the structural signal degrades AUROC to near-random, confirming that the specific entity-relation coverage structure drives detection.

Variant	Overall	Emerging	Novel Ctx
CAGP (full)	.993$\pm$.000	1.00$\pm$.00	1.00$\pm$.00
$\alpha=0$ (cov. only)	.993 $\pm$.000	.975 $\pm$.000	1.00 $\pm$.00
$\alpha=1$ (sem. only)	.500 $\pm$.000	1.00 $\pm$.00	.500 $\pm$.00
Shuffled coverage	.551 $\pm$.016	.593 $\pm$.118	.573 $\pm$.019
Random coverage	.502 $\pm$.004	.809 $\pm$.219	.502 $\pm$.004

Key findings. (1) Removing structural signal ($\alpha=1$, GP only) drops overall AUROC from 0.99 to 0.50—semantic variance alone cannot detect temporal OOD. (2) Shuffling coverage rows (preserving identical row/column sums and density) drops AUROC to 0.55, proving the model exploits *specific* entity-relation associations, not aggregate statistics. (3) Random binary coverage at matched density (1.7%) yields 0.50, ruling out density-driven performance. (4) Coverage alone ($\alpha=0$) matches full CAGP at 0.99 overall, indicating structural (coverage) signal is the primary driver on temporal OOD; semantic signal provides complementary value on emerging-entity slices (0.975 vs 1.00).

C.6 REPRODUCIBILITY: RESULT PROVENANCE

Table 16: Mapping of paper tables to experimental sources. All code is included in the supplementary material.

Table	Dataset	Seeds	Source
1	WN18RR (ours ^a)	42, 123, 456	run_wn18rr_temporal.py
1	WN18RR (baselines ^b)	42, 123, 456	run_wn18rr_missing_baselines.py
1	FB15k-237 (ours ^a)	42, 123, 456	run_wn18rr_temporal.py
1	FB15k-237 (baselines ^b)	42, 123, 456	run_wn18rr_missing_baselines.py
1	YAGO3-10 ^c	42, 123, 456	colab_yago_full.ipynb
1	ICEWS14	42, 123, 456	run_icews14_temporal.py
15	ICEWS14	42, 123, 456	icews14_ablation.py
14	ICEWS14	42, 123, 456	icews14_strict_split.py
2	FB15k-237	42, 123, 456	run_wn18rr_temporal.py
2	WN18RR	42, 123, 456	run_wn18rr_temporal.py
3	All	(same as Table 1)	(same as Table 1)
4	All	(same as Table 1)	(same as Table 1)
5	FB15k-237	42	run_focused_experiments.py

^a UKGE, Energy, U_{sem} , U_{str} , CAGP, RelCondVar. ^b MC Dropout, Deep Ensemble, SNGP. ^c U_{sem} , U_{str} , CAGP only (baselines not evaluated on YAGO).